

Conclusion :-

↳ A^* expands all the nodes that can be reached from the start node on a path where every node on the path has $f(n) < C^*$,
These are surely expanded nodes.

↳ A^* might then expand some of the nodes on the goal contour (where, $f(n) = C^*$) before selecting a goal node.

↳ A^* won't expand any node, $f(n) > C^*$.

↳ A^* with a consistent heuristic is optimally efficient, as it prunes away search tree nodes that are not necessary for finding the optimal path.

1. Complete :- Yes, it's complete.

2. Time :- worst heuristic = ? $\boxed{h(n) = 0} \leq$

$f = g + \cancel{h} \rightarrow A^*$ turns into uniform cost.

$\boxed{\text{time complexity} = O(b^d)}$

3. Space :-

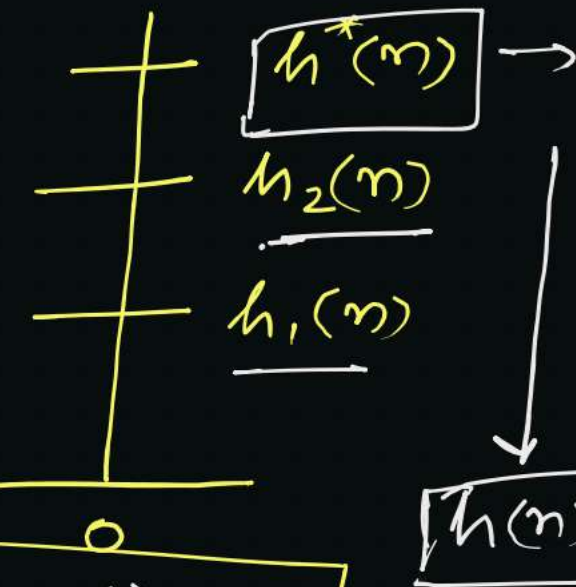
$\boxed{\text{space complexity} = O(b^d)}$

4. Optimal :- Yes, $h(n)$ is admissible.

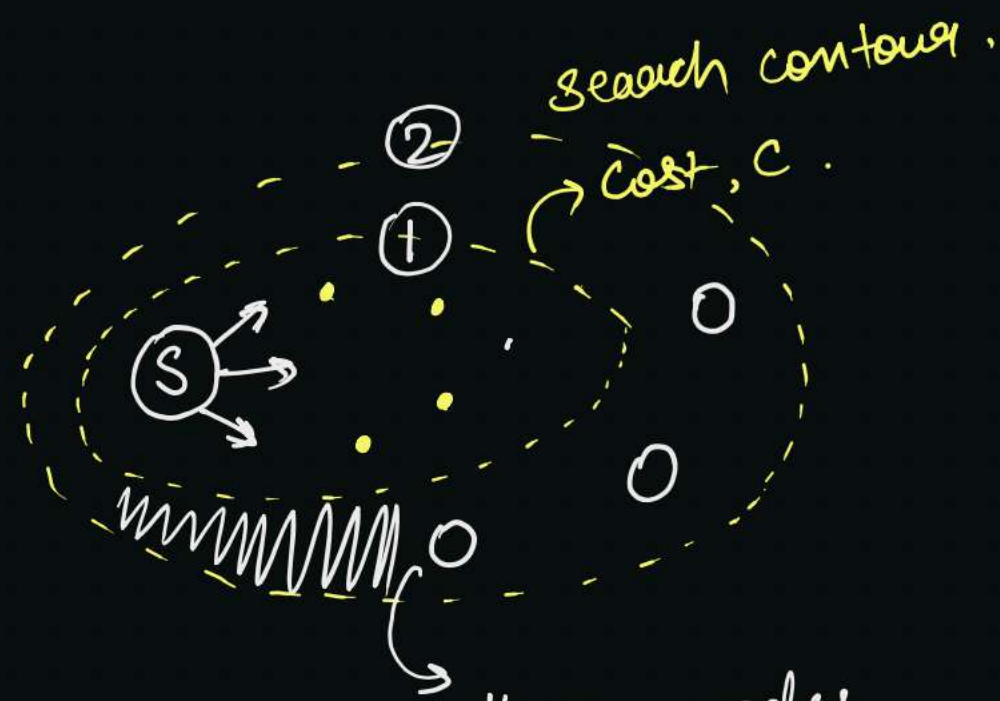
More informed heuristic means less search.

Let A_1 & A_2 use heuristics h_1 & h_2 resp.
Both h_1 & h_2 are admissible.

$\forall(n) (h_2(n) > h_1(n)) \rightarrow h_2 \text{ dominates } h_1$



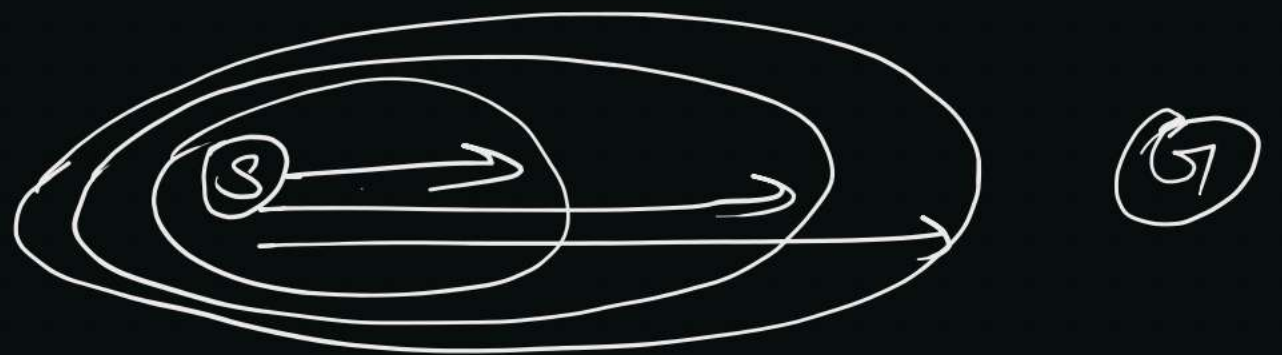
then, $\forall n (\text{expands}(A_2, n) \supset (\text{expands}(A_1, n)))$



$$f = g + h \leq c$$

(G)
$$\left. \begin{aligned} f_1 &= g + h_1 \\ f_2 &= g + h_2 \end{aligned} \right\} h_2 > h_1$$

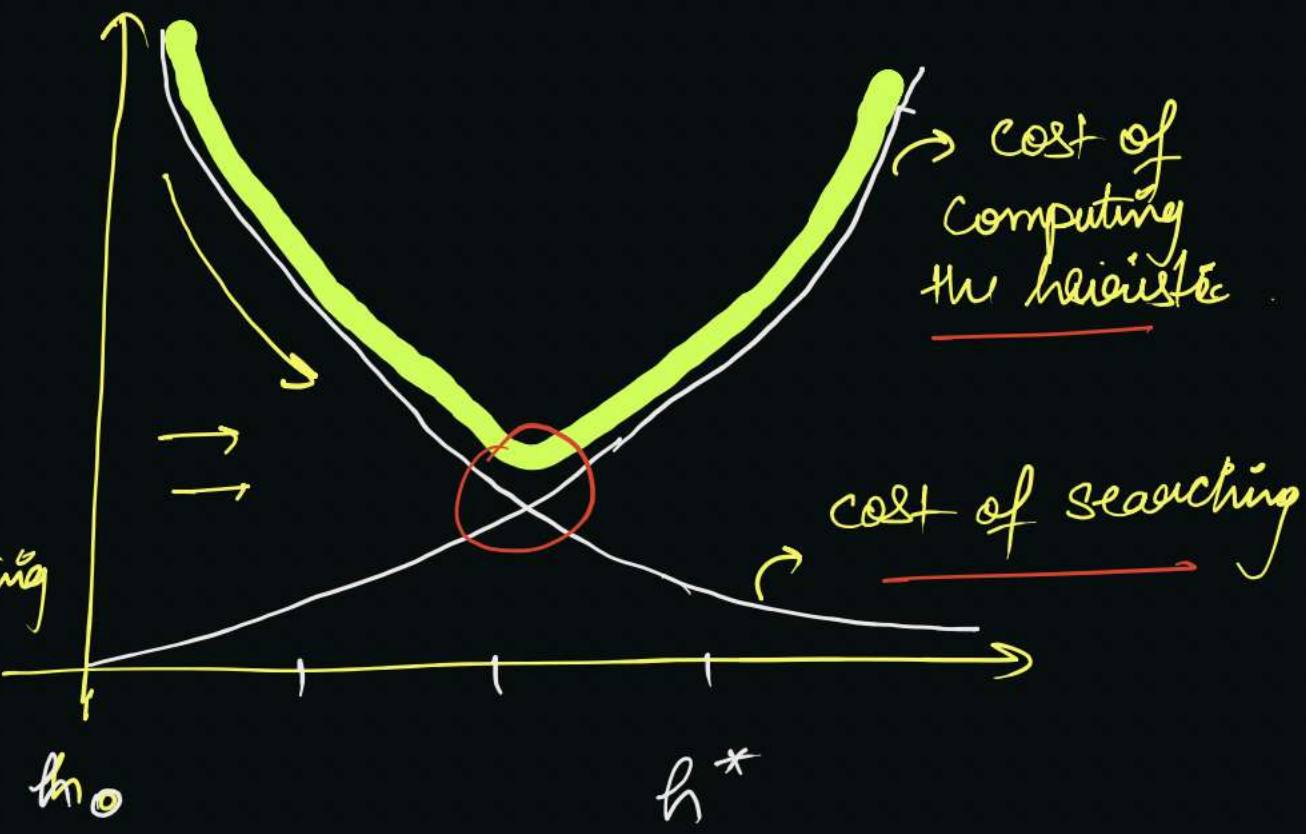
$$f_2 > f_1$$



↳ (more informed)
good heuristic $\rightarrow$ less search

↳ bad heuristic $\rightarrow$ $\begin{cases} \text{good} \\ \text{bad} \end{cases}$ why? more search.

↳ good heuristic $\rightarrow$ $\begin{cases} \text{good} \\ \text{bad} \end{cases} \rightarrow$ cost of computing the heuristic is very high.



not in the scope.

- $\rightarrow$ linear conflicts
- $\rightarrow$ pattern databases. $\rightarrow$
- $\rightarrow$ differential heuristics
- $\rightarrow$ landmarks & shortcuts in heuristics

$$h_1 < h^*$$

$$\boxed{h_2} > \boxed{h^*}$$

$$h_1(n) > h_1(m).$$

$$h^*$$

(S)

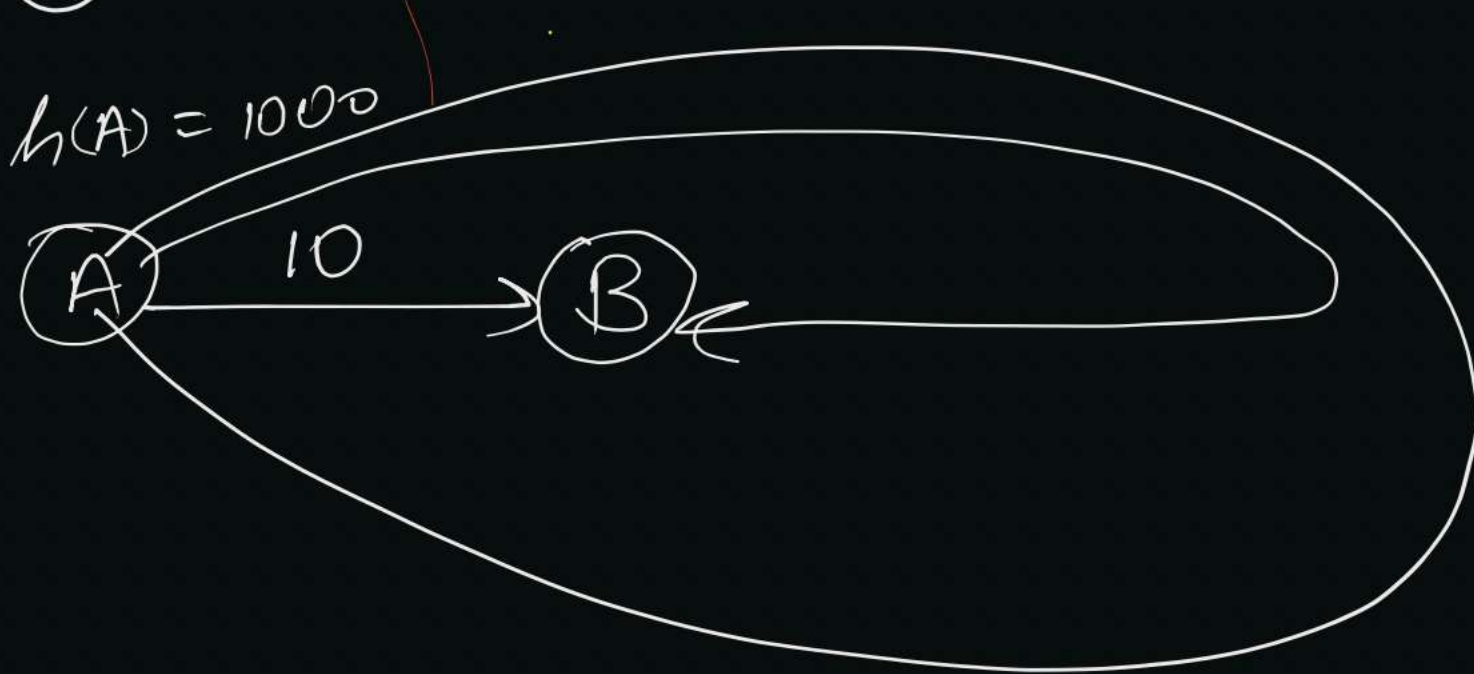
(n)

(m)

(G)

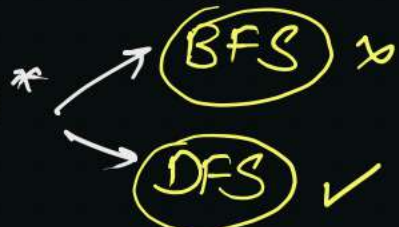
$$f_A = 1000$$

$$h(A) = 1000$$



Memory bound A^* :-

↳ We need to solve the exponential memory problem of A^*



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graph LR; A_star["A*"] --> BFS["BFS ∞"]; A_star --> DFS["DFS ✓"]
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① Beam search :-

- ↳ Limit the size of the frontier (priority queue).
- ↳ Choose a size K , & keep only K nodes with best f -scores.
- ↳ Issue → suboptimality & incomplete.

② Iterative deepening A^* :-

- ↳ we had a limit on depth → uninformed iterative deepening.
- ↳ what parameter we limit → A^* iterative deepening.
 - ↳ we will be putting the limit on f → cost function.

pseudo algorithm :-

limit = ϵ

while (q is not empty)

{

DFS with pruning, remove n , ($f(n)$ > limit)

limit += ϵ

}

performance analysis :-

1. Complete :- Yes.

2. Time :- $O(b^d)$

3. Space :- $O(bd)$

4. Optimality :- Yes.

Weighted A* :-

↳ The pull & the weighted push.

↳ The idea is to weight the heuristic more heavily.

$$f(n) = g(n) + w \cdot h(n) \quad ; \quad w > 1$$

if $w = 1$, $f(n) = g(n) + h(n) \rightarrow A^*$

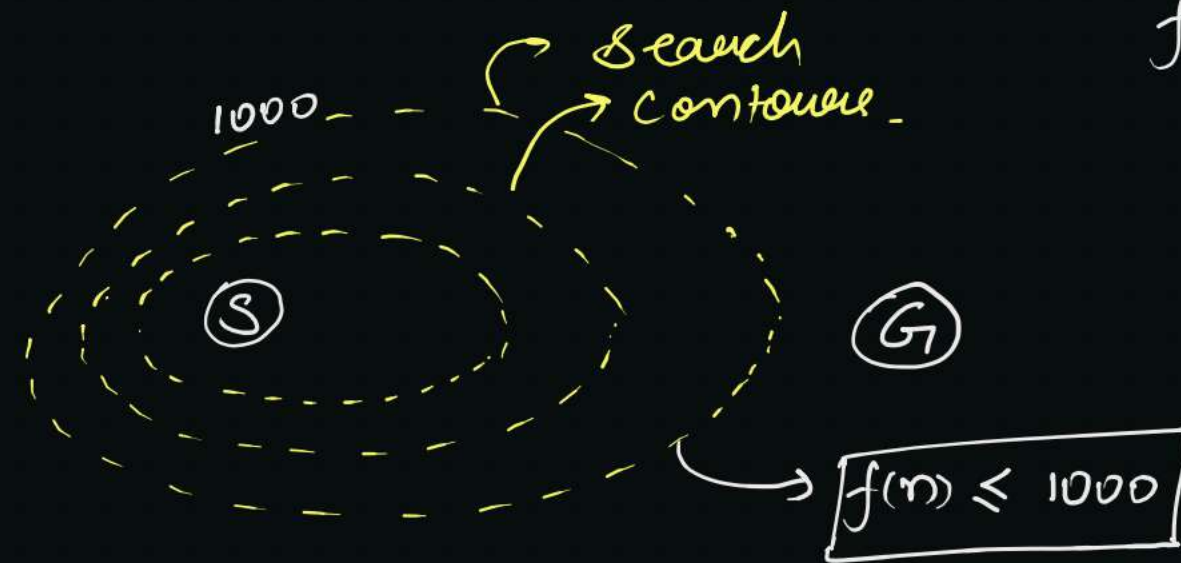
if $w = 0$, $f(n) = g(n) \rightarrow UCS$

if $w = \infty$, $f(n) \propto h(n) \rightarrow$ greedy best first search.

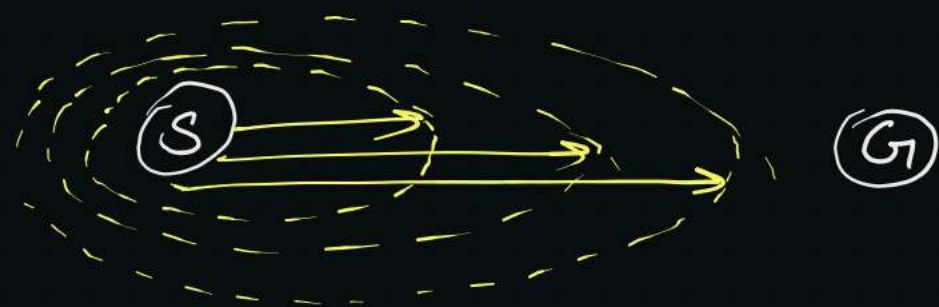
$1 < w < \infty$, $f(n) = g(n) + w \cdot h(n) \rightarrow$ weighted A^* .



uniform cost.



A* search.



weighted A*.

$$\begin{aligned} f(n) &= g(n) + h(n) \\ &= g(n) + w \cdot h(n) \uparrow \uparrow \end{aligned}$$

$$h_n = 10 \uparrow \uparrow$$

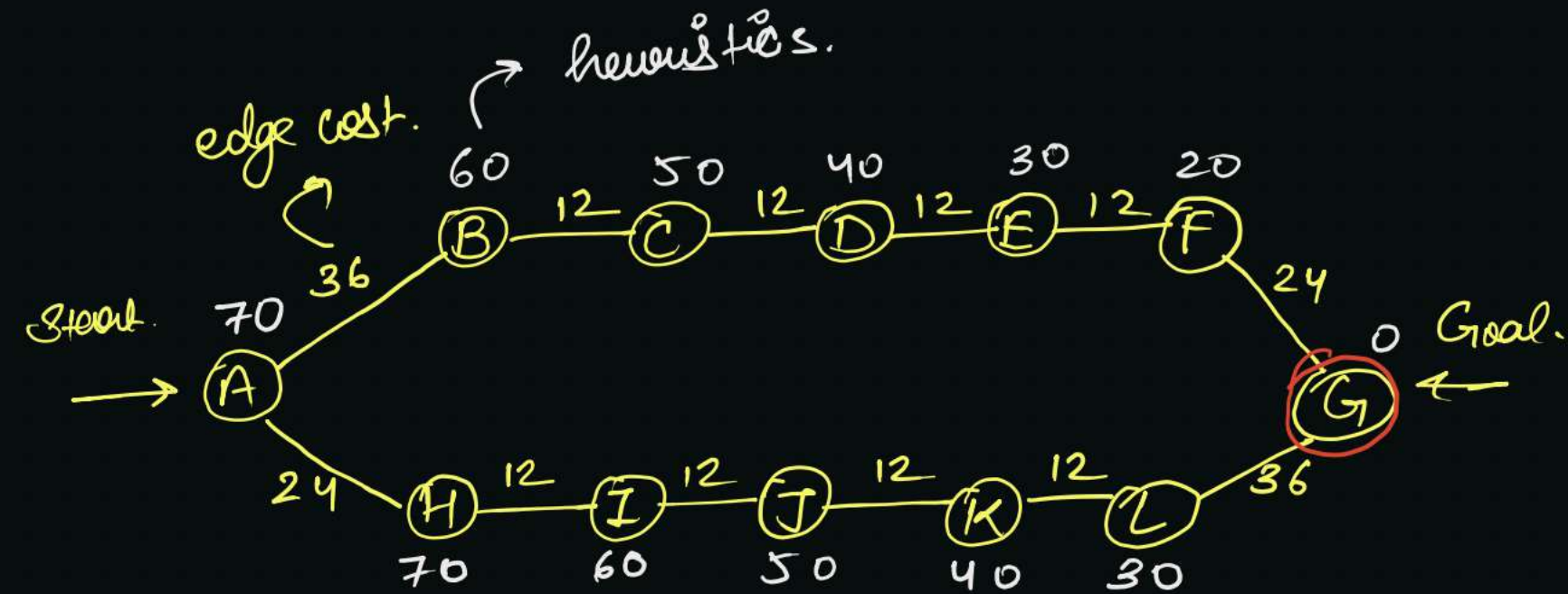
- ↳ In weighted A^* , we can shrink the search contours further but will lose admissibility.
- ↳ Fewer states are explored, so if the optimal path ever strays outside the weighted search's contour, then the optimal path is missed.
- ↳ If the optimal cost of A^* is C^* , then for weighted A_w^* .

cost of solution, $C_w \in [\underline{C^*}, \underline{w \cdot C^*}]$. $\rightarrow$ weight is an non-optimal variation.

$\hookrightarrow$ proof in H.W.

$$\forall(n), \boxed{w \cdot h(n) \not\leq h^*(n)} \times$$

H.W.



- # Find the path $A \rightarrow G$, if
- ① applying A^* search. ?
 - ② applying weighted A^* , $w=2$?

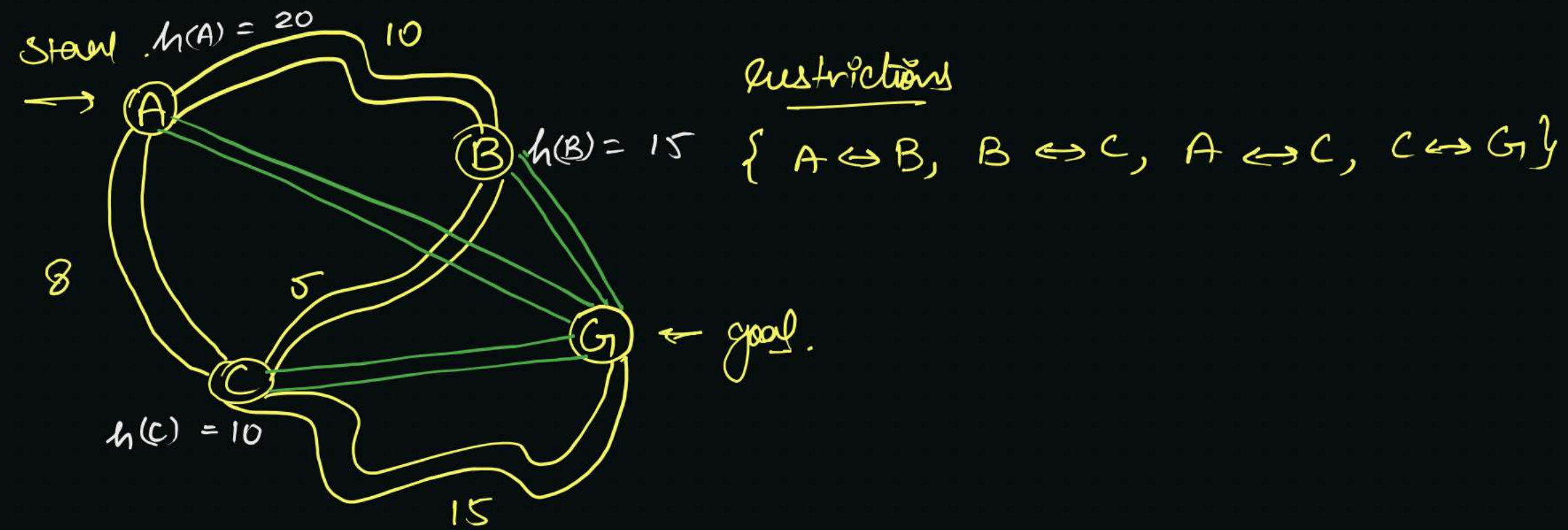
$w = w^*$

$\forall n, w h(n) \leq h^*(n)$

$\checkmark$ always optimal.

Relaxed problems :-

- ↳ In every search problem, we have a set of rules that governs the actions that we take.
- ↳ Relaxing these rules, having fewer restrictions, lead to a relaxed problem.



- ↳ ★ The cost of an optimal solution to a relaxed problem is an admissible heuristic for the original problem.
- ↳ If the paths are relaxed, or a straight road is constructed from city A, B, C towards G, then optimality ↑↑.